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College of Computer Science and Information Technology

ARTI 506 - Data Science and Analytics

Term 2 – 2025/2026 - Assignment 2

Data-Driven Analysis of Tourism Demand and Spending Patterns in Saudi Arabia

AI Year 5

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1.1 Introduction

1.1.1 Introduction

Tourism in Saudi Arabia has become an important contributor to economic growth and development. As part of the Saudi Vision 2030, the tourism sector aims to diversify the economy and promote regional growth. The Kingdom has set a new goal to attract 150 million tourists by 2030 [1]. Understanding tourism patterns is crucial for strategic planning, as tourist demand varies by visitor type (inbound or domestic), destination, and purpose of the visit. A key issue is that the high number of visitors to a particular region does not necessarily mean the highest tourism spending, because spending patterns are influenced by the purpose of the visit. This project aims to address this issue by analyzing tourism datasets of domestic and inbound tourists to uncover patterns in visitor distribution and spending behavior across different regions and visit purposes in Saudi Arabia from 2015 to the first half of 2025. This analysis will provide insights to support the development of data-driven tourism strategies aligned with Saudi Vision 2030.

1.1.2 Problem Statement

Tourism is a significant factor that affects the economy of kingdom of Saudi Arabia. Tourist demand varies by visitor type (inbound or domestic) and by destination and purpose of visit. The main problem is high number of visiting a certain region doesn't mean the highest spending rate because of the purpose of the visits. This gap shows the difficulty in deciding where to focus new investments or how to market other cities differently to increase the tourism movement in Saudi Arabia. This project aims to analyze tourism datasets for both inbound and domestic to show patterns in visitor distribution and spending behavior from 2015-2025H1.

1.1.3 Objectives

This project aims to explore, merge, and analyze four datasets related to the tourism sector in Saudi Arabia. It seeks to provide a comparison between inbound and domestic tourists, generate insights into tourism trends, and forecast future tourism patterns. To predict future tourism, machine learning algorithms will be applied, while exploratory data analysis (EDA) will be conducted to identify trends and patterns within the data. Additionally, data visualization techniques will be utilized to facilitate user interaction with the EDA findings and the predictions generated by the machine learning models.

1.1.4 Dataset description & Justification

For our project, we are using four datasets from the Saudi Open Data Portal that track the development of the Kingdom tourism sector from 2015 to 2025 H1. These datasets include detailed records of inbound and domestic tourists by their main purpose (such as religious, leisure, or business) [2][3], as well as spending and arrival statistics broken down by specific provinces and regions [4][5]. We are using this data to study how tourism trends are changing.
By looking at both visitors number and spending, we can see which trip purposes, like religious or leisure travel, contribute most to the economy and how those patterns differ between local and international tourists.

This project aligns with Saudi Vision 2030 which aims for 150 million visitors and a 10% GDP contribution from tourism by the end of the decade. Using ten years of data, we can apply predictive modeling to forecast demand and identify emerging destinations, providing actionable insights for sustainable planning.

1.1.5 Initial analysis of the dataset's potential

After going through the four tourism datasets from the Saudi Open Data Platform, we believe they have strong potential for our project. The data covers a long period from 2015 to the first half of 2025 and includes important details like visitor type (inbound and domestic), main purpose of visit, region (13 administrative regions), number of tourists, and their spending. This helps us look at tourism from more than one side, not just how many people visit, but also how much they actually spend, which is the main issue we are trying to study. Since the data is organized by year and region, we can easily merge the datasets and calculate useful indicators such as spending per tourist and growth over time. Because the data spans almost ten years, it also allows us to apply forecasting models to predict future tourism trends. Even though the data is yearly and doesn’t show detailed seasonal patterns, it is official, reliable, and detailed enough to support strong exploratory analysis and machine learning models that align with Vision 2030 goals.

1.2 Data Acquisition

1.2.1 Data Loading

In this part, we load the raw data so we could work with it. We used pandas to read the four different CSV files, two about domestic (local) tourism and two about inbound (international) tourism. After loading them, we print the shape of each dataset, which is a quick way to see how many rows and columns we have for each dataset Figure 1, and we use head(3) command to preview the data, which displays the first three rows of each table Figure 2.

```
Domestic by Province : 143 rows, 6 columns
Domestic by Purpose  : 55 rows, 6 columns
Inbound  by Purpose  : 55 rows, 6 columns
Inbound  by Province : 141 rows, 6 columns
```

Figure 1 Dataset shape

--- Domestic by Province ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	DESTINATION PROVINCE NAME EN	Total VISITORS Number	SPENDS by SR
0	DOMESTIC	Overnight	2015	Albaha	1,041,399	1,119,776,018.55
1	DOMESTIC	Overnight	2015	Alqassim	946,993	832,038,134.23
2	DOMESTIC	Overnight	2015	Aseer	5,091,717	6,482,489,011.94

--- Domestic by Purpose ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	VISIT PURPOSE GROUPED EN	VISITORS	SPENDS
0	DOMESTIC	Overnight	2015	Business	1,859,752	3,247,666,761.42
1	DOMESTIC	Overnight	2015	Leisure	19,181,792	21,183,455,240.20
2	DOMESTIC	Overnight	2015	Other	2,335,328	4,059,351,265.38

--- Inbound by Purpose ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	VISIT PURPOSE GROUPED EN	VISITORS	SPENDS
0	INBOUND	Overnight	2015	Business	1,880,575	10,090,826,436.41
1	INBOUND	Overnight	2015	Leisure	1,560,665	4,954,068,163.96
2	INBOUND	Overnight	2015	Other	2,392,366	9,067,180,026.90

--- Inbound by Province ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	DESTINATION PROVINCE NAME EN	VISITORS	SPENDS
0	INBOUND	Overnight	2015	Albaha	6,092	42,822,952.65
1	INBOUND	Overnight	2015	Alqassim	41,902	143,586,806.97
2	INBOUND	Overnight	2015	Aseer	59,257	503,026,385.00

Figure 2 Preview first 3 rows of each dataset

1.2.2 Data Summary

Once the data was loaded, we wanted to see how it was structured. We use the .info() function to get a high-level summary Figure 3. This showed us the names of every column and, more importantly, the data types helping us realize that many numbers were currently being treated as objects instead of actual numbers, which we will clean later.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 143 entries, 0 to 142
Data columns (total 6 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   TOURIST TYPE EN             143 non-null    object
 1   TRIP TYPE                   143 non-null    object
 2   YEAR                        143 non-null    object
 3   DESTINATION PROVINCE NAME EN 143 non-null    object
 4   Total VISITORS Number       143 non-null    object
 5   SPENDS by SR                143 non-null    object
dtypes: object(6)
memory usage: 6.8+ KB


<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55 entries, 0 to 54
Data columns (total 6 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   TOURIST TYPE EN          55 non-null     object
 1   TRIP TYPE                55 non-null     object
 2   YEAR                     55 non-null     object
 3   VISIT PURPOSE GROUPED EN 55 non-null     object
 4   VISITORS                 55 non-null     object
 5   SPENDS                   55 non-null     object
dtypes: object(6)
memory usage: 2.7+ KB


<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55 entries, 0 to 54
Data columns (total 6 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   TOURIST TYPE EN          55 non-null     object
 1   TRIP TYPE                55 non-null     object
 2   YEAR                     55 non-null     object
 3   VISIT PURPOSE GROUPED EN 55 non-null     object
 4   VISITORS                 55 non-null     object
 5   SPENDS                   55 non-null     object
dtypes: object(6)
memory usage: 2.7+ KB


<class 'pandas.core.frame.DataFrame'>
RangeIndex: 141 entries, 0 to 140
Data columns (total 6 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   TOURIST TYPE EN             141 non-null    object
 1   TRIP TYPE                   141 non-null    object
 2   YEAR                        141 non-null    object
 3   DESTINATION PROVINCE NAME EN 141 non-null    object
 4   VISITORS                    141 non-null    object
 5   SPENDS                      141 non-null    object
dtypes: object(6)
memory usage: 6.7+ KB
```

Figure 3 Datasets information

1.2.3 Statistical Description

To understand the data better, we use the .describe() command. This gave us a mathematical preview of the data Figure 4, showing things like the count of unique values and the most frequent entries. Since our numbers were stored as text, it helped us identify things like the range of years (2015 to 2025 H1) and the specific regions included in the study.

--- Domestic Province ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	DESTINATION PROVINCE NAME EN	Total VISITORS Number	SPENDS by SR
count	143	143	143	143	143	143
unique	1	1	11	13	143	143
top	DOMESTIC	Overnight	2015	Albaha	1,041,399	1,119,776,018.55
freq	143	143	13	11	1	1

--- Domestic Purpose ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	VISIT PURPOSE GROUPED EN	VISITORS	SPENDS
count	55	55	55	55	55	55
unique	1	1	11	5	55	55
top	DOMESTIC	Overnight	2015	Business	1,859,752	3,247,666,761.42
freq	55	55	5	11	1	1

--- Inbound Purpose ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	VISIT PURPOSE GROUPED EN	VISITORS	SPENDS
count	55	55	55	55	55	55
unique	1	1	11	5	55	55
top	INBOUND	Overnight	2015	Business	1,880,575	10,090,826,436.41
freq	55	55	5	11	1	1

--- Inbound Province ---

	TOURIST TYPE EN	TRIP TYPE	YEAR	DESTINATION PROVINCE NAME EN	VISITORS	SPENDS
count	141	141	141	141	141	141
unique	1	1	11	13	141	141
top	INBOUND	Overnight	2015	Alqassim	6,092	42,822,952.65
freq	141	141	13	11	1	1

Figure 4 Description of each dataset

1.3 Data Cleaning

1.3.1 Cleaning Function

Instead of manually fixing every dataset one by one, we wrote a reusable cleaning function to automate the process. This function does three main things, it removes commas from numbers so the computer can do math with them, renames columns so they match across all four files, and converts the year column into a numeric format, for example, turning "2025 H1" into 2025.5.

```
--- Domestic Province ---
  Shape      : 143 rows x 6 columns
  Columns    : ['TOURIST TYPE EN', 'YEAR', 'DESTINATION PROVINCE NAME EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
  Year range  : 2015 – 2025 H1

--- Inbound Province ---
  Shape      : 141 rows x 6 columns
  Columns    : ['TOURIST TYPE EN', 'YEAR', 'DESTINATION PROVINCE NAME EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
  Year range  : 2015 – 2025 H1

--- Domestic Purpose ---
  Shape      : 55 rows x 6 columns
  Columns    : ['TOURIST TYPE EN', 'YEAR', 'VISIT PURPOSE GROUPED EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
  Year range  : 2015 – 2025 H1

--- Inbound Purpose ---
  Shape      : 55 rows x 6 columns
  Columns    : ['TOURIST TYPE EN', 'YEAR', 'VISIT PURPOSE GROUPED EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
  Year range  : 2015 – 2025 H1
```

Figure 5 Dataset cleaning function

1.3.2 Handling Missing Values

Before starting any deep analysis, we check for values in our data where information might be missing. We scanned every row and column in all four datasets for empty values. The result in Figure 6 showed 0 Missing Values for every file, which confirmed that our datasets were complete and ready for analysis.

```
--- Domestic (Province) ---
Missing Values: 0
------------------------------
--- Inbound (Province) ---
Missing Values: 0
------------------------------
--- Domestic (Purpose) ---
Missing Values: 0
------------------------------
--- Inbound (Purpose) ---
Missing Values: 0
------------------------------
```

Figure 6 Number of missing values

1.3.3 Outlier Identification

In this step, we look for extreme values that are much higher or lower than the rest. We use boxplots to visualize these outliers in Figure 7. We discovered that while there are extreme values, like very high visitor counts in Makkah and Riyadh, they aren't mistakes, they just represent the busiest regions. To stop these huge numbers from making our charts look messy, we capped them at the 95th percentile, which keeps the data honest without letting the extreme value hide the smaller details.

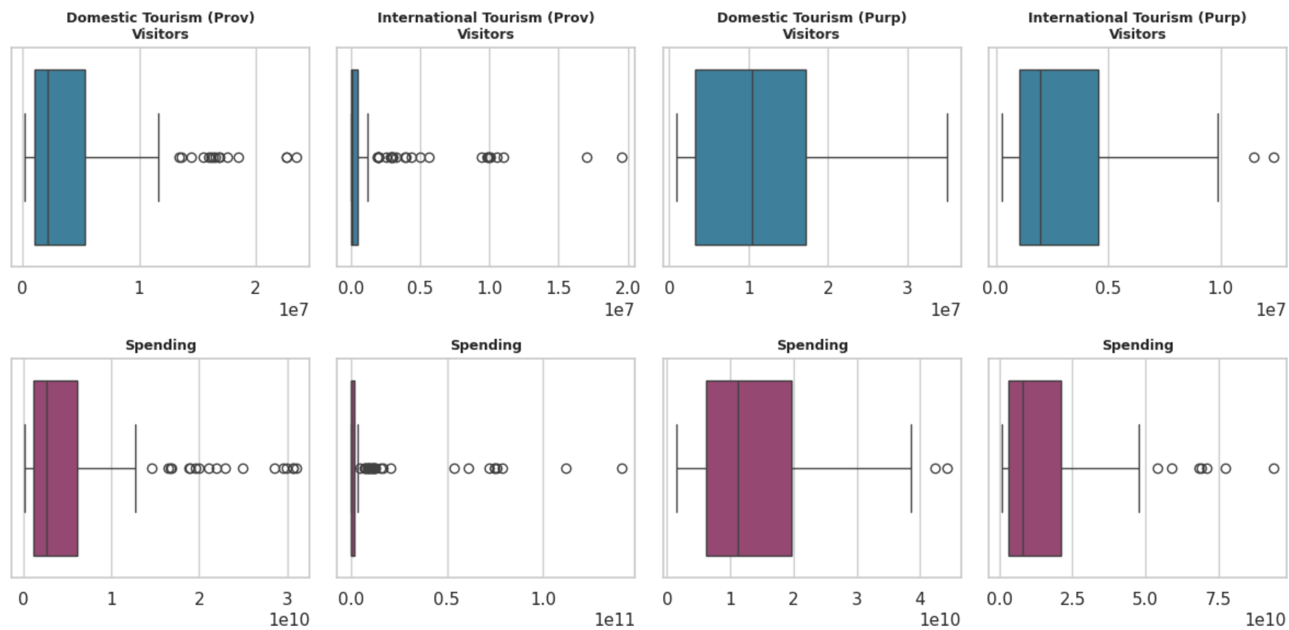

Figure 7 Outliers identification

1.4 Feature Engineering

1.4.1 Adding New Features

Utilizing existing data to generate new features is an essential step in data science and machine learning, as it enhances the ability of the model to capture useful patterns. In this project, a new feature, SPEND_PER_TOURIST, was created and added to all four datasets (Domestic Province, Inbound Province, Domestic Purpose, and Inbound Purpose). The feature can be derived by dividing SPENDS over VISITORS, it provides a measure of the average spending per tourist and allow to understand the individual spending behavior across different provinces and purposes, rather than relying on total spends. Figure 8 presents sample values from the Domestic Purpose dataset.

	YEAR	VISIT_PURPOSE_GROUPED_EN	VISITORS	SPENDS	SPEND_PER_TOURIST
0	2015	Business	1859752.0	3.247667e+09	1746.290237
1	2015	Leisure	19181792.0	2.118346e+10	1104.352254
2	2015	Other	2335328.0	4.059351e+09	1738.236027
3	2015	Religious	11972476.0	1.015693e+10	848.356278
4	2015	VFR	11101085.0	9.771687e+09	880.246154

Figure 8 Sample values of 'SPEND_PER_TOURIST' feature.

1.4.2 Visualizing skewness and applying Log Transformation

In the outlier identification stage, Figure 7 showed a clear right skewness in both visitor counts and total spending across all four datasets. This skewness resulted from several factors, such as Makkah having higher visitor numbers compared to other provinces, and inbound tourists spending more on specific purposes like religious and business travel. Skewed data can impact analysis and machine learning model performance negatively, and it may cause bias in the results due to extreme values. To address this issue, a log transformation (log(x + 1)) was applied to both VISITORS and SPENDS across all datasets, and one was added to handle zero values. As shown in Figure 9, the original distribution of visitor data shows right skewness. After applying the log transformation, the distribution becomes balanced and close to a normal (bell-shaped) distribution. These transformed features will be utilized in later stages of the project as it suitable for machine learning models.

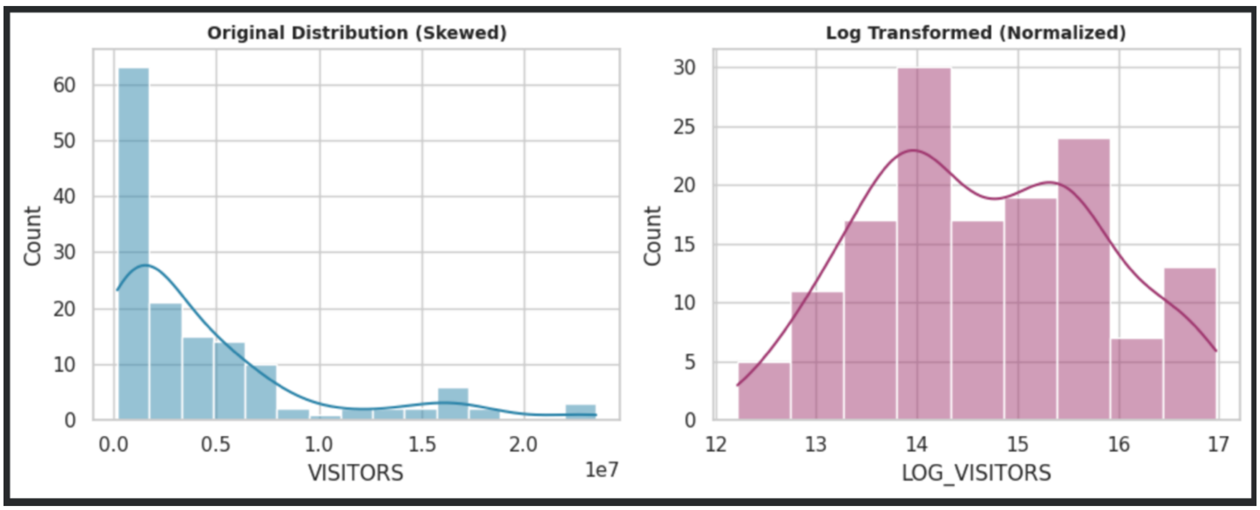

Figure 9: Visualization of skewed distribution vs. normal distribution.

1.5 Exploratory Data Analysis

1.5.1 Overall Tourism Growth Trends (2015–2025)

The Figure 10 illustrates the total number of tourists in Saudi Arabia from 2015 to 2025 H1, distinguishing between domestic tourism and international tourism. The x-axis represents the timeline, while the y-axis shows the total number of visitors. From the graph, we can see that domestic tourism consistently exceeds international tourism throughout the period. Both segments dropped sharply in 2020 due to the impact of COVID-19, followed by a strong recovery from 2021 onward, with tourism levels peaking around 2024 before a slight drop in 2025 H1. The 2025 H1 figure is lower simply because it only covers half the year. International tourism shows greater fluctuations compared to the more stable domestic trend.

Figure 10 Line chart showing total tourism growth in Saudi Arabia from 2015 to 2025 (H1)

1.5.2 Regional Visitor Trends Over Time

The chart in Figure 11 tracks how visitor numbers changed over time in the five most visited provinces in Saudi Arabia, from 2015 to 2025 H1. Makkah consistently attracts the highest number of tourists, followed by Riyadh and the Eastern Province with stable growth patterns, while Madinah and Aseer maintain lower but steady visitor numbers. The COVID-19 impact in 2020 is visible across all five regions, but recovery speed and peak levels differ, suggesting that regional tourism drivers are not uniform and that targeted regional strategies would be more effective than a one-size-fits-all approach.

Figure 11 multi-line chart illustrating visitor trends over time for the top five regions in Saudi Arabia

1.5.3 Visitor Volume vs. Spending Shares (Market Share Analysis)

These two pie charts in Figure 12 compare the percentage of total visitors and total spending that comes from domestic versus international tourists. Domestic tourists make up 77.6% of all visitors but contribute only 44.7% of total spending. International tourists are only 22.4% of visitors but generate 55.3% of spending. This highlights the importance of attracting international tourists as a key driver of tourism revenue, while domestic tourism remains essential for maintaining overall visitor numbers and market stability.

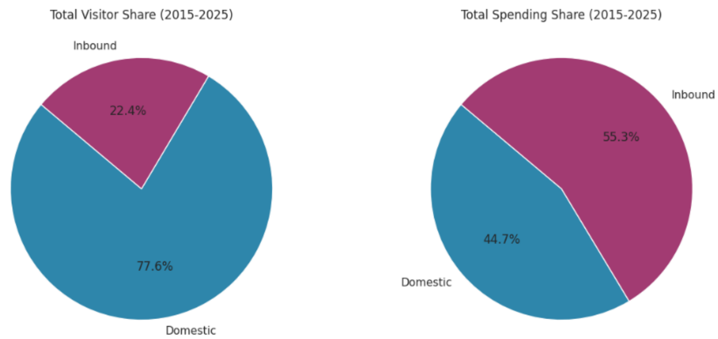

Figure 12 Pie charts showing visitor and spending shares for domestic and inbound tourism in Saudi Arabia

1.5.4 Comparing Average Spending per Tourist

Figure 13 shows the trend in average tourist spending using SAR for both domestic and international tourists from 2015 to 2025. International tourists consistently spend more than domestic tourists, with values fluctuating but generally increasing over time. In contrast, domestic tourist spending remains relatively stable with only slight variations. Both dipped during COVID, but international spend per tourist has reached its highest level ever at around 6,300 SAR in 2025 H1, while domestic spend per tourist has remained largely flat between 1,000 and 1,500 SAR throughout the entire period. Overall, International tourists contribute higher spending per visitor, while domestic spending shows steady but lower levels.

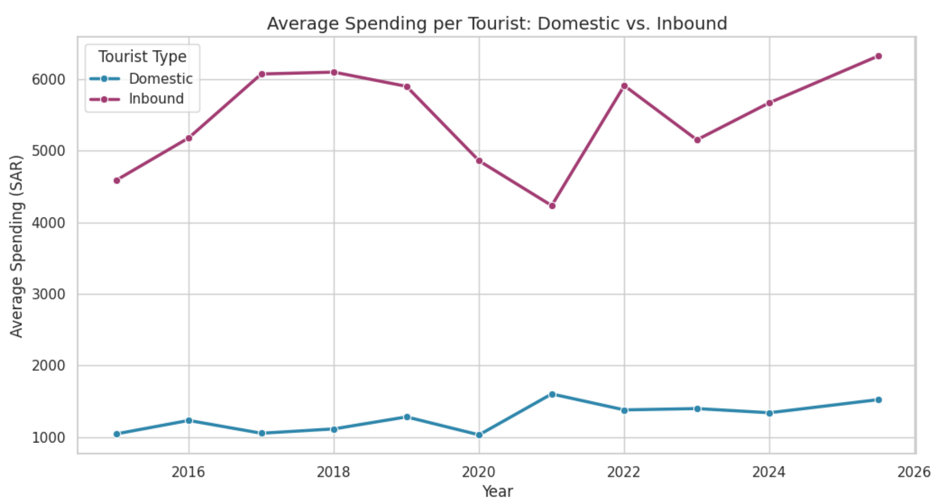

Figure 13 Line chart showing the average spending per tourist (SAR) for domestic and inbound tourists from 2015 to 2025.

1.5.5 Comparing Inbound vs. Domestic Travel Purposes

The Bar chart in Figure 14 compares the number of domestic and international visitors by purpose of visiting. Leisure is the most common purpose for domestic tourists, followed by religious visits, then VFR (visiting friends and relatives), which is a personal trip type associated with lower spending, helping explain why domestic tourists spend less per person. On the other hand, international tourists are highest in religious visits, with more evenly spread across the remaining purposes compared to domestic tourists.

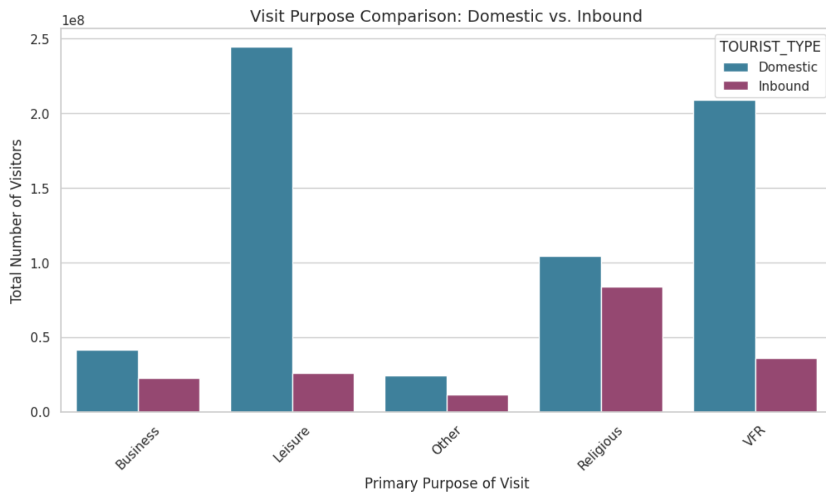

Figure 14 Bar chart comparing domestic and inbound visitors by primary purpose of visit

1.5.6 Tourism by Destination Comparison (Regional Performance)

This chart in Figure 15 shows which provinces attract the most tourists, comparing domestic and international visitors side by side across all 13 regions. Makkah has the highest number of visitors in both categories. For domestic tourism, Riyadh and the Eastern Province also have high visitor numbers, followed by Madinah and Aseer, while other provinces attract significantly fewer tourists. For international tourists, Makkah alone accounts for most arrivals, confirming that international tourism is heavily driven by religious visits to the holy cities.

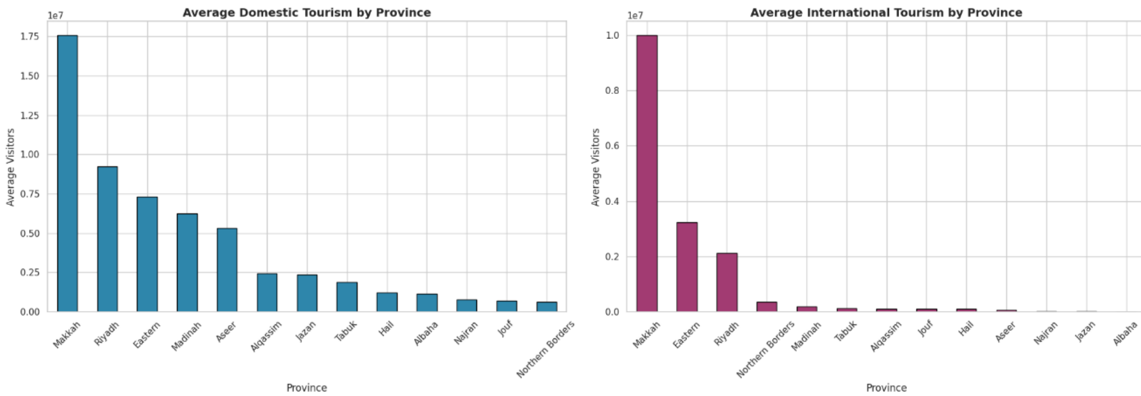

Figure 15 Bar charts illustrating the average number of domestic and international tourists by province.

1.6. Advanced Insights & Correlation

1.6.1 Feature Selection

Feature selection is an essential step in data analysis and machine learning, where the most relevant features are selected to improve model performance and reduce the impact of irrelevant variables. In this project, seven key features were selected based on their importance in representing tourism patterns. One of the primary features is TOURIST_TYPE, which distinguishes between domestic and inbound tourists. Since the datasets are structured based on both provinces and purposes, DESTINATION_PROVINCE_NAME_EN and VISIT_PURPOSE_GROUPED_EN were also included. The feature YEAR_NUM represents the temporal dimension of the data, containing 11 unique values spanning from 2015 to the first half of 2025. This wide time range enables the analysis of trends over time and supports forecasting future tourism patterns. Additionally, log-transformed features, LOG_VISITORS and LOG_SPENDS, were computed to reduce data skewness and improve model stability. These transformations help machine learning models better capture patterns. Finally, SPEND_PER_TOURIST was included as a derived feature to represent individual spending behavior. This feature provides insights into how much each tourist spends across different regions and purposes, enhancing the ability of the model to analyze patterns in tourism.

1.6.2 Apply Correlation

In this section, Pearson correlation is utilized to measure and analyze the relationships between different numerical features across the four datasets. The correlation coefficient ranges from -1 to +1, where a value close to +1 indicates a strong positive relationship between two features, while a value close to -1 indicates a strong negative relationship. A value near 0 suggests that there is no linear relationship between the variables [6].

1.6.2.1 Relationship Between Visitor Volume and Total Spending

This section aims to examine the relationship between visitor volume and total tourism spending. A scatter plot was used to visualize the relationship between the number of visitors and total spending for both domestic and inbound tourism datasets. In addition, a linear trend line was applied to identify the overall direction of the relationship, along with Pearson correlation to quantify its strength. As shown in Figure 16, the results indicate a strong positive relationship between visitor numbers and total spending for both domestic and inbound tourists, with a correlation coefficient of 0.97. This suggests that increases in visitor volume are strongly associated with higher tourism revenue. However, the data points are scattered around the trend line, indicating that visitor numbers alone do not fully determine total spending. Some provinces may attract a high number of visitors while generating lower spending per tourist. This supports the idea that visitor quantity and economic contribution are not always directly proportional.

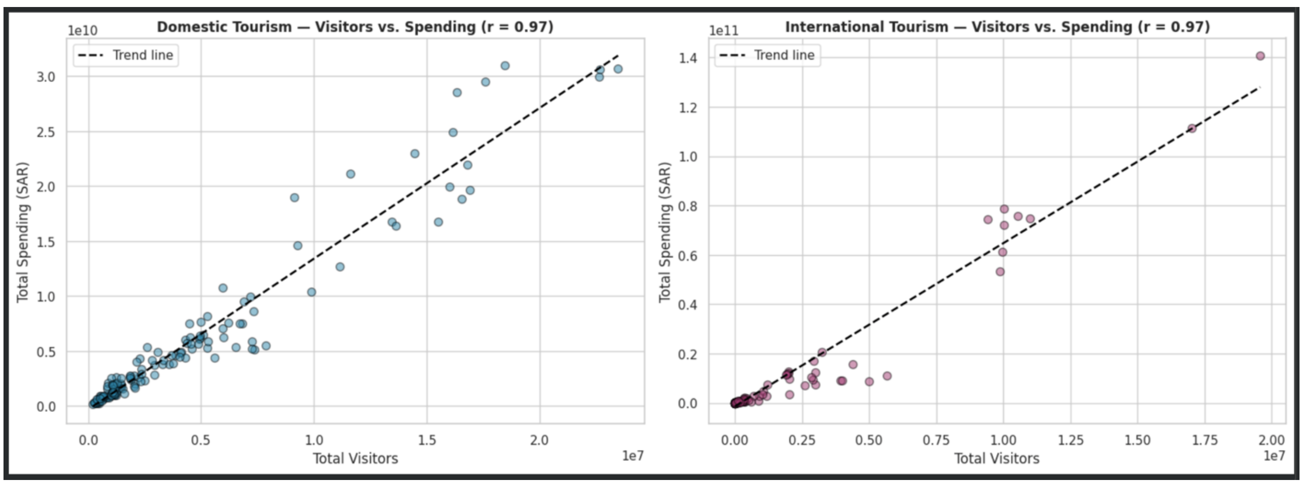

Figure 16: Scatterplot of visitors vs. spending for both domestic and inbound tourism.

1.6.2.2 Feature Correlation Heatmaps Across All Datasets

This section examines the relationships between the main numerical features across the four tourism datasets using correlation heatmaps. The analysis focuses on YEAR_NUM, VISITORS, SPENDS, and SPEND_PER_TOURIST features. Correlation matrices were computed and visualized using heatmaps as shown in Figure 17, where red indicates a strong positive relationship, blue represents a negative relationship, and lighter colors indicate weak or no relationship. Across all datasets, a strong positive correlation was observed between VISITORS and SPENDS, with values ranging between 0.96 and 0.97. This indicates that higher visitor volumes are strongly associated with increased total tourism spending. The relationship between VISITORS and SPEND_PER_TOURIST varies across datasets. In most cases, the correlation is weak to moderate. However, an exception appears in the Domestic Tourism by Purpose dataset, where a negative correlation of -0.42 is observed. This suggests that the purposes attracting the highest number of domestic tourists tend to generate lower spending per individual. Similarly, the relationship between SPENDS and SPEND_PER_TOURIST differs by tourism type. In inbound tourism by purpose, a relatively strong positive correlation of 0.64 is observed, indicating that higher total spending is associated with increased spending per tourist. In contrast, domestic tourism by purpose shows a weak negative relationship of -0.20, suggesting a different spending pattern. Finally, the relationship between YEAR_NUM and SPEND_PER_TOURIST shows a moderate positive trend in domestic tourism, indicating a gradual increase in spending per tourist over time. In contrast, inbound tourism exhibits only a weak relationship, suggesting that spending behavior among international tourists remains relatively stable over the years.

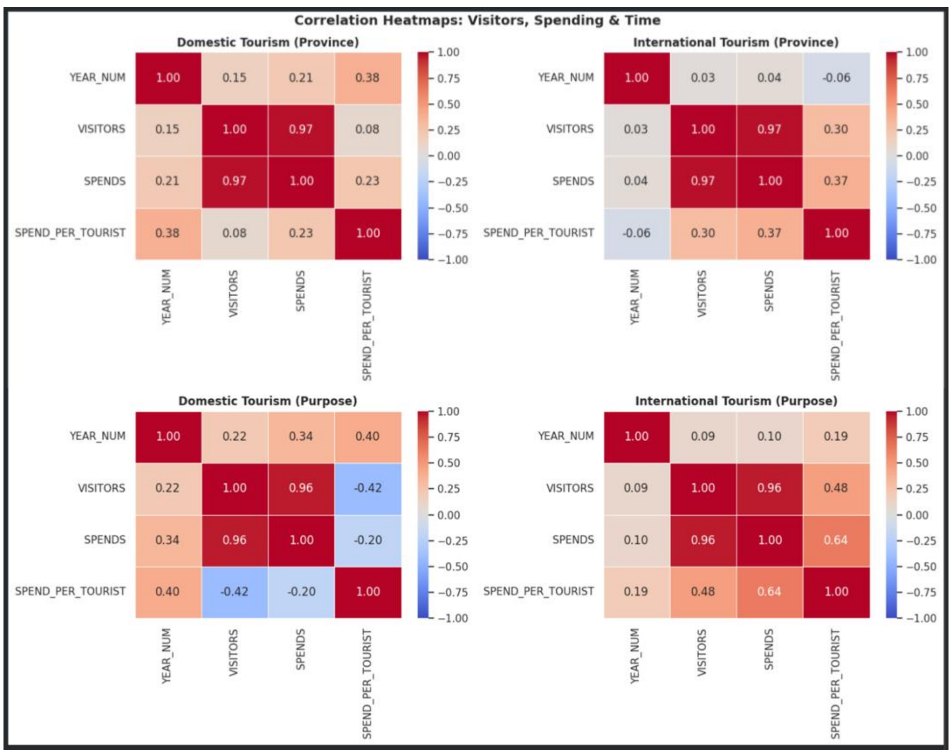

Figure 17: Correlation heatmaps across all datasets for visitors, spending, and time.

1.6.2.3 Pearson Correlation Summary

This section provides a summary of the correlation results identified across all four datasets. A summary table was created to recognize the key relationships between the four selected numerical features, offering a clearer comparison of correlation patterns observed in the heatmaps. The table categorizes correlation using color coding by identifying strong correlations (≥ 0.7) in green, moderate correlations (≥ 0.4) in yellow, and weak correlations are left plain. As shown in Figure 18, VISITORS and SPENDS show the strongest correlation across all datasets, confirming that tourism revenue is dependent on the number of visitors. In contrast, features associated with time (YEAR_NUM) show weak correlations, indicating that tourism growth does not follow a linear trend over the years and is influenced by external events such as COVID in 2020. An exception is observed in the relationship between SPEND_PER_TOURIST and YEAR_NUM for domestic tourism by purpose, where a moderate correlation suggests a gradual increase in individual spending over time.

Dataset	Visitors vs Spending	Visitors vs Year	Spending vs Year	Spend/Tourist vs Year
Domestic Tourism (Province)	**0.970000**	0.150000	0.210000	0.380000
International Tourism (Province)	**0.970000**	0.030000	0.040000	-0.060000
Domestic Tourism (Purpose)	**0.960000**	0.220000	0.340000	0.400000
International Tourism (Purpose)	**0.960000**	0.090000	0.100000	0.190000

Figure 18: Pearson correlation summary table.

2. Chapter 2:

2.1 Data preprocessing summary

The data preprocessing stage is important to prepare the dataset and ensure the data is clean, consistent, and suitable for analysis and modeling. We applied several preprocessing step techniques to improve the data quality and enhance model performance. We started by loading the data using **pd.read_csv()** to read the datasets, previewing the first 3 rows of each dataset using **data.head()**, explore and understand the data using **data.info()** and **data.describe()**.

We developed a cleaning function to facilitate consistent cleaning across all datasets. It contains three main tasks: remove commas from numbers to enable mathematical operations, rename columns to standardize across all datasets to facilitate merging and unified analysis, and convert the year column to a numeric format. Then, we checked for missing values across all datasets, and the result showed that there are no missing values in all datasets.

After that, outliers were identified using boxplot visualization, especially in visitors and spending. These extreme values were mainly observed in regions that have high tourism activity, like Makkah and Riyadh. But since these values represent the actual real-world patterns rather than errors, we didn't remove them. Instead, we handled the outliers by capping at the 95^{th} percentile. This ensures that the data is not influenced by the huge numbers and preserves the overall distribution of data.

In addition, we apply a log transformation ($\log(x + 1)$) to VISITORS and SPENDS across all datasets to solve the skewed distribution that is influenced by the outliers. Then we add one to handle zero values. This transformation helps to normalize the data, reduce skewness, and make the data suitable for the machine learning models.

Finally, we implemented a categorical encoding to convert the categorical variables into numerical values using pd.**get_dummies()** to make sure the data is ready for the machine learning model.

2.2 Model Selection

The modeling phase of this research focuses on predicting tourism spending per visitor and understanding the factors that influence tourism demand, in order to support the strategic goals of Saudi Vision 2030.

Given the insights from the Exploratory Data Analysis (EDA), which revealed a complex relationship between visitor types, regional destinations, and primary purposes of visit, a multi-model approach was adopted. To achieve a balance between statistical interpretability and predictive performance, two models were selected:

- Linear Regression (LR).
- Random Forest (RF).

2.2.1 Linear Regression (LR): The Statistical Baseline

Linear Regression serves as the foundational benchmark for this study, chosen primarily due to the strong overall trend identified between tourism activity and spending patterns.

2.2.1.1 Theoretical Justification:

The Pearson correlation analysis conducted in the preliminary stage indicated a strong positive relationship between tourism-related variables, suggesting that a linear model can capture the general direction of spending behavior. Linear Regression is widely used for modeling relationships between dependent and independent variables and provides a simple yet effective baseline for predictive tasks [7].

2.2.1.2 Predictive Utility:

LR is effective in quantifying the contribution of numerical features, such as log-transformed visitor counts (LOG_VISITORS), while also incorporating categorical variables like destination and visit purpose, in relation to the target variable LOG_SPENDS_PER_VISITOR.

2.2.1.3 Interpretability:

In an academic and policy-making context, Linear Regression provides clear and interpretable coefficients that make it easier to understand how each factor contributes to tourism spending. This transparency is important for explaining results and supporting data-driven decision-making [7].

2.2.2 Random Forest (RF): Robust Non-Linear Modeling

While Linear Regression captures the overall trend, the tourism dataset contains significant variations across regions and visit purposes that require a more flexible modeling approach.

2.2.2.1 Handling Categorical Complexity:

The dataset includes multiple destination regions and visits purposes (e.g., Religious, Leisure, Business, VFR), each associated with different spending behaviors. Random Forest, introduced by Breiman (2001), is an ensemble learning method based on multiple decision trees that can naturally handle complex and non-linear relationships in the data [8].

2.2.2.2 Resistance to Variability and Noise:

Tourism data often contains variability due to seasonal trends, regional differences, and fluctuations in visitor behavior. Random Forest uses a bagging (bootstrap aggregating) technique, which helps reduce variance and improves the stability of predictions, making it suitable for real-world datasets [8].

2.2.2.3 Feature Interaction:

Feature interaction occurs when the effect of one variable on the target depends on another variable. While linear models require these interactions to be explicitly defined, Random Forest captures them automatically through its tree-based structure.

Mechanism of Automated Interaction:

- In a decision tree, the model performs recursive splitting; for example, it may first divide the data based on destination region and then further split it based on visit purpose. This allows the model to learn that spending behavior differs depending on combinations of factors.

Application to Saudi Tourism Patterns:

- The EDA results showed that spending is not determined by visitor volume alone, but also depends heavily on where tourists go and why they visit. For instance, certain destinations associated with religious or business travel tend to generate higher spending per visitor compared to others.

Refining the "Spend per Tourist" Insight:

- Since the LOG_SPENDS_PER_VISITOR feature was used as the target variable, Random Forest helps capture the variation in individual spending behavior across different segments. This makes it more effective in identifying cases where high visitor numbers do not necessarily translate into high economic contribution.

2.2.3 Summary of Model Selection

In this study, Linear Regression and Random Forest were selected to complement each other. Linear Regression provides a simple and interpretable baseline model that captures the general trend in tourism spending, while Random Forest offers a more flexible approach capable of modeling complex patterns and interactions in the data.

Using both models allows for a more comprehensive analysis and enables performance comparison, helping determine whether tourism spending behavior follows a linear pattern or requires more advanced modeling techniques for accurate prediction.

2.3 Results & Model Evaluation

2.3.1: Linear Regression model results

For Linear regression, we've calculated the Impact "coefficient", which acts as a mathematical weight used to predict the spending per visitor. Each value shows the specific amount in SAR added or subtracted from the baseline average. This metric shows how much the province or purpose of visiting might independently influence the overall spending behavior.

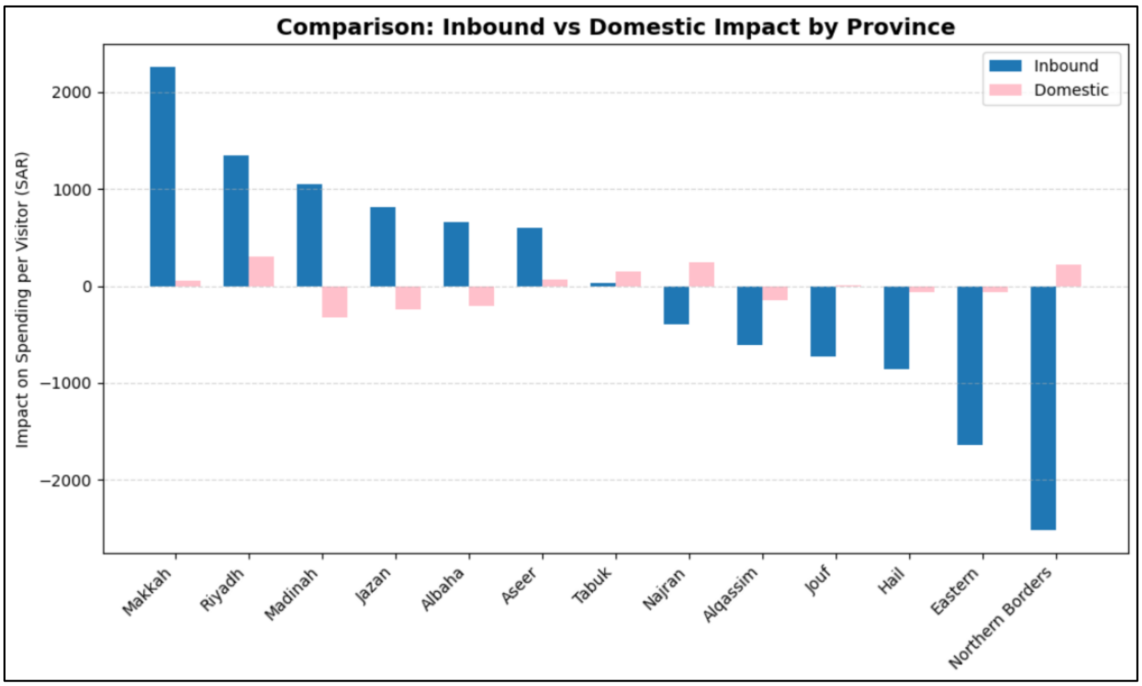

Figure 19 Comparison: Inbound vs Domestic Impact by Province

The bar chart in Figure 19 shows a strong divergence in spending patterns between inbound and domestic spending across Saudi provinces. For inbound tourists, Makkah and Riyadh were the highest positive contributors, increasing predicted spending by more than 2,000 SAR and 1,300 SAR, respectively, while regions like “Northern Borders” and “Eastern province” showed negative impacts, dropping nearly to -2,500 SAR. However, Domestic tourism shows a very different view. Domestic tourists' impact was consistently near the zero line, in contrast to the inbound tourists, which reflects a consistent and stable spending pattern among the tourists, regardless of the destination. What was noticeable was the gap in "Makkah", the impact difference between domestic and inbound tourists. Inbound tourists spend significantly more than domestic tourists, which might indicate that domestic tourists have careful spending habits.

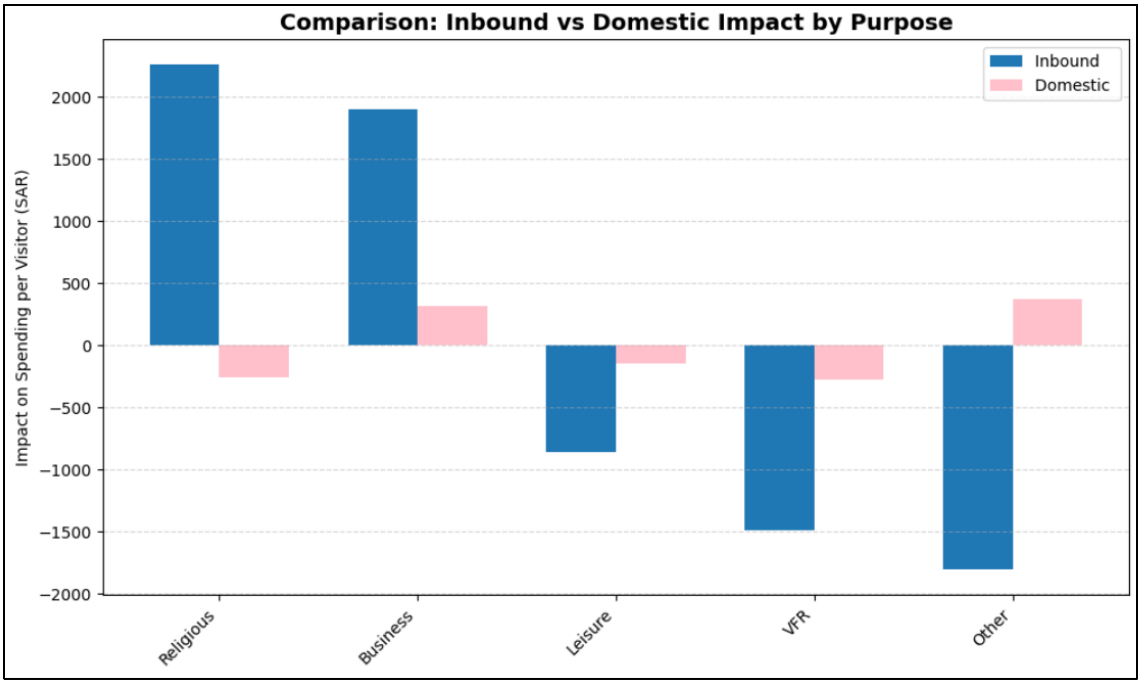

Figure 20 Comparison: Inbound vs Domestic Impact by Purpose

Here, we did a comparison of the impact on spending per visitor between inbound vs domestic tourists based on the purpose of the visit. The bar chart in Figure 20 illustrates a noticeable difference between the two types of tourists. For the inbound tourists, Religious and business have the most positive impact, increasing predicted spending by approximately 2,200 SAR and 1,900 SAR. In contrast, purposes such as VFR (visiting friends and relatives) and others had the lowest negative impact, dropping nearly -1,500 to -1,800 SAR. While domestic tourism still shows stability, with most purposes lying nearly on the zero line. Purposes like "Business" and "Other" show a minor positive impact between 0 SAR and 500 SAR. Similar to inbound tourism, "VFR" had the most negative impact on domestic tourism. Notably and conversely to inbound tourism, "Religious" had a negative impact on domestic tourism by less than 0 SAR.

```
Inbound Purpose LR Results
R2 Score on Test Set: 78.55%
Avg Error (MAE): 736.02
Squared avg Error (MSE): 703033.14
Square root avg Error (RMSE): 838.47

  Domestic Purpose LR Results
R2 Score on Test Set: 60.07%
Avg Error (MAE): 242.46
Squared avg Error (MSE): 115398.39
Square root avg Error (RMSE): 339.70

  Inbound province LR Results
R2 Score on Test Set: 52.64%
Avg Error (MAE): 863.71
Squared avg Error (MSE): 1072504.89
Square root avg Error (RMSE): 1035.62

  Domestic province LR Results
R2 Score on Test Set: 17.22%
Avg Error (MAE): 245.62
Squared avg Error (MSE): 86527.53
Square root avg Error (RMSE): 294.16
```

Figure 21 LR Performance metrics

As shown in Figure 21, The results show the performance metrics for the Linear Regression model based on province and purpose of visit. Showing that the purpose of the visit significantly scores a higher prediction than the destination province. For inbound tourists, the purpose of visit achieved the highest reliability with an R2 score of 78.55% and moderate error levels, which indicates that the purpose of the visit is a main factor of international tourists' spending behavior. Similarly, the domestic purpose results show an R2 score with 60.07% with lower error performance, which indicates that local tourism has a predictable spending behavior. When comparing the province-based tourists results, the inbound province result achieved lower predictive power with an R2 by 52.64% and higher error performance, which reflects that the location alone is not sufficient to determine the international tourists' spending pattern. The domestic province scored the lowest R2 score with 17.22% while having lower error levels, which indicates that location plays a limited factor in explaining the local tourists' spending behavior.

2.3.2 Random Forest (RF) model results

For the Random Forest model, we've calculated the feature importance "weight" to evaluate how much each variable contributes to the model's prediction accuracy. Unlike the impact coefficient, the importance value represents the predictive power, not a specific SAR value. This metric shows which provinces or purposes of visit the model will prioritize, as it is crucial to determine the overall spending behavior.

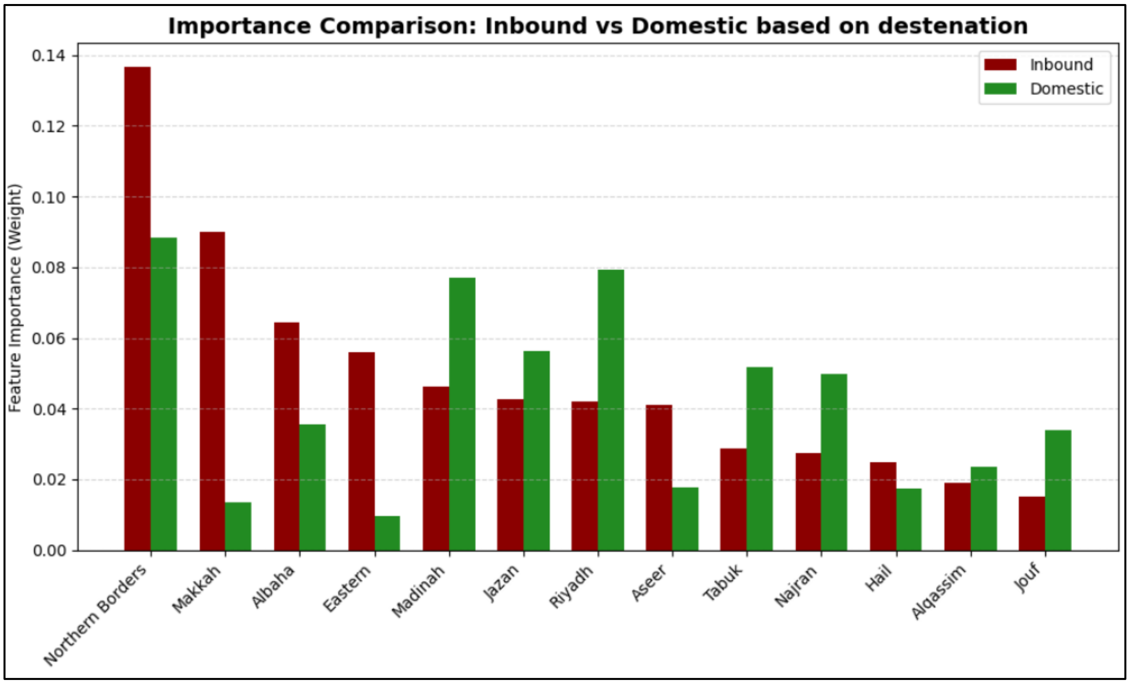

Figure 22 Importance Comparison: Inbound vs Domestic based on province

In Figure 22, The bar chart shows the Importance comparison between Inbound vs Domestic based on destination. The chart shows how each province contributes to the accuracy of the model prediction, which shows that not all provinces contribute equally. For inbound tourists, the model relies heavily on "Northern Borders" and "Makkah" between 0.08 and 0.14, representing them as the strongest signals for inbound spending variations. In contrast, Domestic tourism focuses more on "Riyadh" and "Madinah", which carry greater importance compared to other provinces, with weights between 0.06 and 0.08.

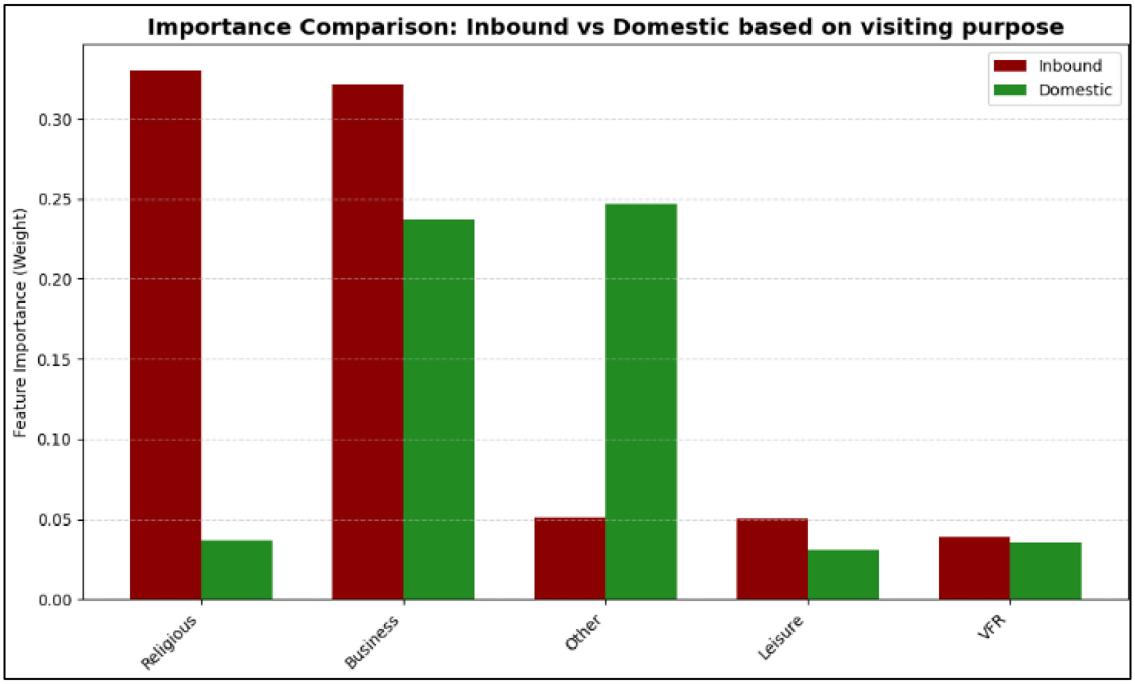

Figure 23 Importance Comparison: Inbound vs Domestic based on visiting purpose

In Figure 23, the bar chart shows the Importance comparison between Inbound vs Domestic based on visiting purposes. The chart shows how each purpose contributes to the accuracy of the model prediction, which shows that not all provinces contribute equally. For inbound tourists, the model relies heavily on "Religious" and "Business" between 0.30 and 0.35, representing them as the strongest signals for inbound spending variations. In contrast, Domestic tourism focuses more on "Business" and "Other”, which carry greater importance compared to other provinces, with weights between 0.20 and 0.25.

```
Inbound Purpose RF Results
R2 Score on Test Set: 82.88%
Avg Error (MAE): 562.81
Squared avg Error (MSE): 561148.53
Square root avg Error (RMSE): 749.10

 Domestic Purpose RF Results
R2 Score on Test Set: 57.60%
Avg Error (MAE): 224.35
Squared avg Error (MSE): 122555.40
Square root avg Error (RMSE): 350.08

  Inbound province RF Results
R2 Score on Test Set: 49.06%
Avg Error (MAE): 827.39
Squared avg Error (MSE): 1153509.99
Square root avg Error (RMSE): 1074.02

  Domestic province RF Results
R2 Score on Test Set: 58.84%
Avg Error (MAE): 166.67
Squared avg Error (MSE): 43026.66
Square root avg Error (RMSE): 207.43
```

Figure 24 RF Performance Metrics

As shown in Figure 24 The results show the performance metrics for the Random Forest model based on province and purpose of visit. It shows that for the inbound tourists, the purpose of visit achieved the highest predictive performance with an R2 score of 82.88% with an MAE of 562.81. In contrast, the Domestic purpose shows lower predictive power with an R2 score achieved 57.60%, but it also achieved lower errors, showing more stable and less variable local spending. Also, by comparing the province-based data, the inbound tourists' data have the lowest R2 score with 49.06%, with significantly higher error, which indicates that location is not sufficient on its own to explain the spending behavior of international tourists. On the other hand, the domestic tourists show a slightly higher R2 score with 58.84%, with low error performance, which shows that local tourists are more influenced by the provinces.

2.3.3 Comparing the Models results

The performance of Linear Regression and Random Forest was evaluated across multiple tourism categories using R^2, MAE, RMSE, and MSE to assess both model fit and prediction accuracy.

The R^2 comparison shows how well each model explains the variability in the data, as shown in Figure 25. Random Forest achieved higher R^2 values in inbound tourism by purpose (0.83 vs 0.79) and domestic tourism by province (0.59 vs 0.17), indicating better model fit in these categories. In contrast, Linear Regression performed slightly better in domestic tourism by purpose (0.60 vs 0.58) and inbound tourism by province (0.53 vs 0.49), suggesting more linear relationships in these cases.

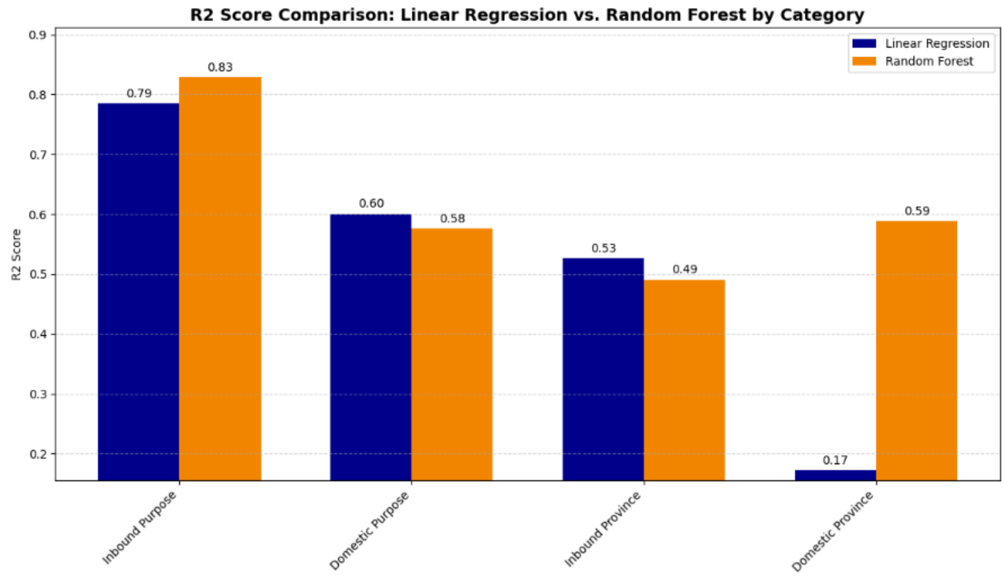

Figure 25: Comparison of R² scores between Linear Regression and Random Forest across all categories.

The MAE results show the average absolute prediction error for each model, as presented in Figure 26. Random Forest achieved lower MAE in inbound tourism by purpose (562.81 vs 736.02), inbound tourism by province (827.39 vs 863.71), and domestic tourism by province (166.67 vs 245.62), indicating more consistent predictions. Although Linear Regression achieved a slightly higher R^2 in domestic tourism by purpose, Random Forest still produced lower MAE (224.35 vs 242.46), indicating more accurate predictions.

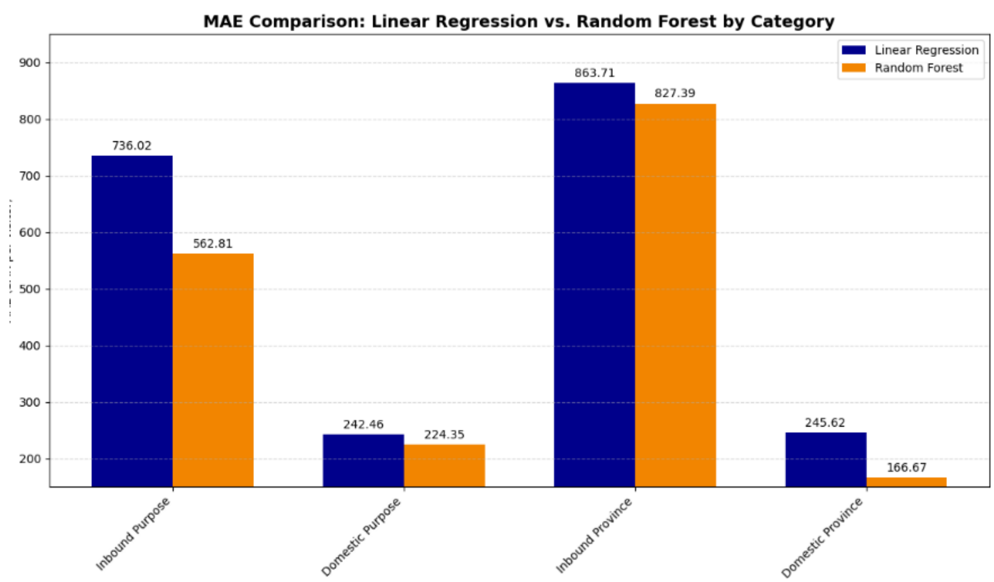

Figure 26: Comparison of MAE values between Linear Regression and Random Forest across all categories.

The RMSE results compare the prediction error of both models, as illustrated in Figure 27, where lower values indicate better performance. Random Forest achieved lower RMSE in inbound tourism by purpose (749.10 vs 838.47) and domestic tourism by province (207.43 vs 294.16), indicating more accurate predictions. However, Linear Regression showed slightly lower RMSE in domestic tourism by purpose (339.70 vs 350.08) and inbound tourism by province (1035.62 vs 1074.02).

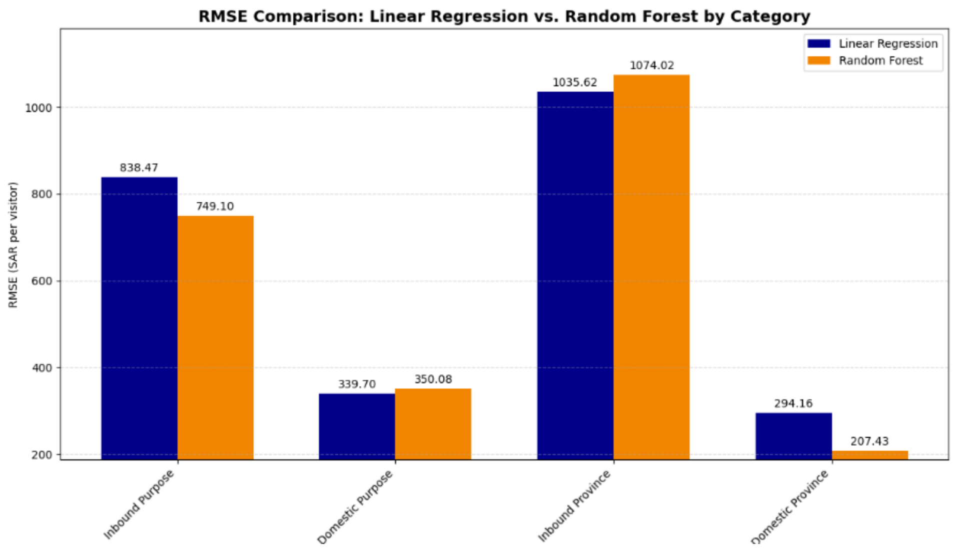

Figure 27: Comparison of RMSE values between Linear Regression and Random Forest across all categories.

The MSE comparison further highlights differences in prediction error, as shown in Figure 28. Random Forest achieved lower MSE in inbound tourism by purpose (561148.53 vs 703033.14) and domestic tourism by province (43026.66 vs 86527.53). On the other hand, Linear Regression performed better in domestic tourism by purpose (115398.39 vs 122555.40) and inbound tourism by province (1072504.89 vs 1153509.99), confirming similar patterns observed in RMSE.

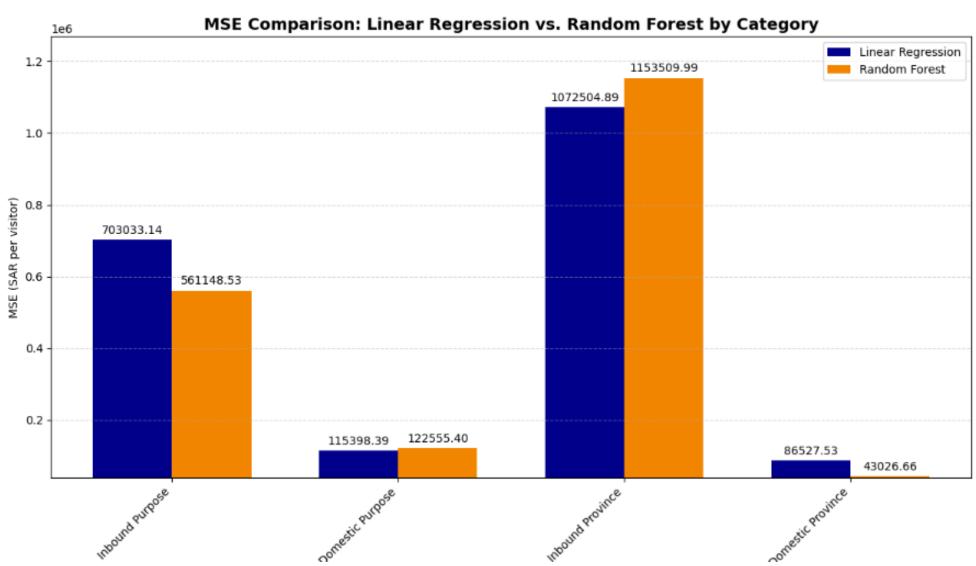

Figure 28: MSE comparison confirming overall error trends between the models.

Overall, the results show that Random Forest generally performs better in capturing complex patterns, especially in domestic tourism by province (R^2 = 0.59 vs 0.17), while Linear Regression performs competitively in simpler or more linear cases such as domestic tourism by purpose (R^2 = 0.60 vs 0.58). This highlights the importance of selecting models based on the nature of the data.

2.4 Discussion and Analysis

2.4.1 Key Findings

The study aims to analyze and predict tourism spending per visitor and identify the main factors (purpose and destination) that influence the spending behavior in Saudi Arabia. The results show that both visit purpose and destination affect spending, but their impact differs clearly between inbound and domestic tourists.

The findings indicate that visit purpose is a stronger factor than destination, especially for inbound tourists. This means that ‘why tourists travel’ has a greater effect on spending than ‘where they go.’ This can be explained by the nature of different travel purposes. For example, religious and business trips often involve higher fixed costs such as accommodation, transportation, and other services, which directly increase spending behavior. In contrast, destination alone does not always determine how much a tourist spends, since visitors may choose different spending levels even within the same location.

In addition, inbound tourists show higher and more variable spending compared to domestic tourists. This illustrates that international visitors contribute more to high-value tourism, as they typically spend on flights, hotels, and other services. However, domestic tourists show more stable spending patterns with small differences across purposes and destinations. This stability indicates that local visitors tend to spend in a more consistent and controlled way.

Finally, the comparison between models shows that tourism spending cannot be fully explained by simple linear relationships. While Linear Regression helps explain the direction of impact (increase or decrease in spending), it struggles when patterns become more complex. Random Forest, on the other hand, captures the complex relationship. This explains why it performs better in scenarios where spending behavior is more variable, particularly for domestic tourism by destination.

2.4.2 Role of Visit Purpose in Spending Behavior

Visit purpose plays a significant role in influencing spending per visitor, with clear differences between inbound and domestic tourists.

Inbound tourists spend significantly more when travelling for Religious and Business purposes. This is because these types of trips usually involve higher costs, such as flights, accommodation, and related services. On the other hand, Leisure and VFR are linked to lower spending, since tourists are more likely to manage their spending, such as staying with family instead of booking hotels.

For domestic tourists, the effect of visit purposes is less strong. Although Business and Other purposes show slightly higher spending, the difference between all purposes is relatively small. This means that domestic tourists tend to spend in a more consistent and controlled way, regardless of their reason for travel.

Overall, the visit purpose has a stronger influence on inbound tourists, while domestic tourists show more stable spending across different purposes.

2.4.3 Role of Destination in Spending Behavior

Destination also plays an important role in explaining spending patterns, with noticeable differences between inbound and domestic tourists. But its impact is different from visit purposes.

For the inbound tourists, spending is higher in a few key provinces, such as those associated with religious and economic activities. This means that certain locations attract higher-spending visitors, such as Makkah and Riyadh. In contrast, for domestic tourists, spending is more evenly distributed across different provinces. The differences between locations are smaller, which shows that domestic tourists travel does not depend on a specific destination for their spending.

This suggests that destinations alone are not enough to fully explain spending behavior. However, for domestic tourists, destinations become more useful when using the Random Forest model, which captures more complex patterns. This indicates that location still plays a role, but in a less direct and more complex way compared to the visit purpose.

Overall, destinations have an influence on spending, but it's less important than the visit purpose, particularly for inbound tourism.

2.5 Limitations

Although the Saudi tourism datasets proved to be useful, a number of limitations were noted:

1. Limited time detail: The datasets are annual and the year 2025 is only represented as 2025 H1. This restricts the capability to examine monthly or seasonal trends like Hajj, Ramadan or summer tourism.

2. Aggregated data: Data is aggregated based on province, purpose and type of tourist. It does not encompass in-depth tourist profiles, including nationality, age, duration of stay, type of accommodation and travel behavior.

3. Partial 2025 data: As 2025 would only cover half-year data in 2025, it cannot be directly compared with half-year data on previous years.

4. External factors not included: It is possible that tourism spending is sensitive to inflation, events, hotel prices, and travel costs but these variables were not included in the datasets.

5. Outlier influence: There are locations that are characterized by extremely high values of visitor and spending. These are not inaccuracies but they can have a very powerful influence on the analysis and the model results.

2.6 Challenges and Solutions

- Handling numeric columns: It imported some columns like VISITORS and SPENDS as text due to comma separators. The answer was to delete commas and to transform them into numeric data types.
- Combining multiple datasets: The project used four separate datasets for domestic and inbound tourism by province and purpose. The remedy was to normalize the names of columns and combine related datasets to facilitate analysis.
- Sorting the year column: The value 2025 H1 had sorting problems as it is not a normal numeric year. The answer was to have a numeric year column with 2025 H1 being 2025.5.
- Managing outliers: There were some provinces with very large values of visitor and spending. The remedy was to truncate the extreme values and to transform the data as a logarithm to reduce skew without making the data meaningless.
- Comparing domestic and inbound tourism: the number of visitors was higher in the domestic tourism whereas the amount of money spent by a tourist was a lot higher in the inbound tourism. The remedy was to compare both indicators of the number of visitors and their spending, including spending per tourist and market share.
- Dashboard implementation : To construct the Power BI dashboard, it was necessary to create combined tables, DAX measures, slicers, and sorted year fields. The answer was to lay out the dashboard into individual pages with an overview, domestic tourism, inbound tourism, visit purpose, and final findings.

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Appendix

GitHub Link

https://github.com/faihadiln-spec/saudi-tourism-demand-spending-analysis.git

Power BI Dashboard Link

https://udksa-my.sharepoint.com/:u:/g/personal/2220002599_iau_edu_sa/IQDJOpGZsvo3RKmA8OiSYsNAAeFQlxNuG0M8B-rNYS5A7QI?e=hahiMM